Writing Research Proposals

A practical reference based on the lecture · The goal is to increase your chances of writing a convincing, feasible proposal.

The fundamental law of academic writing

The story of your research is not the story of your proposal.

1. Start with your audience, not your topic

Ask: Who will review this? What do they want to hear? What would make them feel their time (and funding) is well spent? Reviewers are not automatically invested in your research. Make your case from their perspective.

For a student grant, the proposal should help reviewers believe that:

- You can finish in a reasonable time.
- You will carry out good research, potentially leading to publications.
- The funding will be money well spent.

2. Show that the topic matters to research

An appealing idea is not enough. Explain why the problem matters to the literature: What open challenge does it address? Why are current approaches insufficient? Why is progress here scientifically interesting? A topic can be important to humanity without yet being a persuasive research proposal.

3. Show that you can actually do it

Reviewers also need evidence of feasibility and of your ability to execute. Use relevant coursework, technical skills, prior projects, a prototype, a preliminary result, or expertise you already have. Select evidence that supports the proposed work instead of listing every achievement.

4. Build a credible bridge

Your previous work and your intended research may be far apart. The proposal should connect them through a concrete, plausible research direction. You may need to reshape how you present the project so that the connection becomes visible. The key is to show a coherent path, not to reproduce the exact sequence of how you arrived at the idea.

Helpful question: “Given what I have already done, why is this proposed next step both interesting and believable?”

Turning the principles into a proposal

A writing workflow and a final submission check

5. Start from something you already have

Before writing the grand plan, do a little of the work. Start with a paper in progress, a prototype, an example, a first definition, or a preliminary result. This gives you a real foothold from which to describe the next steps. A research plan is like a map: it is much harder to draw a reliable one if you have not walked any part of the terrain.

6. Remember the medium: one story across many materials

Your application may contain a proposal, recommendation letters, a CV, statements, evidence of progress, or a presentation. Coordinate these pieces. Each should reinforce the same credible research trajectory, but each can play a different role. Focus effort on the pieces you control, while coordinating the others wherever possible.

7. Use generative AI without outsourcing your thinking

Writing is thinking. If you outsource your writing, you outsource your thinking. AI can help refine language and structure, but you must fully understand and own the argument: the research problem, the scientific gap, the methods, and why the plan is plausible. Do not submit a proposal you cannot explain and defend yourself.

Before you submit: a quick self-check

- Audience: Do I know what my reviewers are trying to assess?
- Importance: Have I explained the gap in the literature, not just why my idea sounds impressive?
- Feasibility: Have I shown evidence that I can do the work?
- Coherence: Is there a clear bridge from what I know/have done to what I propose?
- Concrete starting point: Can I point to some initial research, prototype, example, or result?
- Consistency: Do the proposal, CV, letters, and other application elements support the same story?
- Ownership: Could I confidently defend every major claim without relying on AI-generated phrasing?

Examples to revisit in the lecture slides

Scientific importance versus a merely ambitious idea (slides 20–28); demonstrating readiness through relevant experience (slides 29–36); constructing a bridge from past work to a new direction (slides 38–47); starting from a real piece of work (slides 49–52).

Bottom line: A good proposal does more than describe an exciting project. It persuades a particular audience that the research matters, that the plan is credible, and that you are prepared to carry it out.